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1  "include">#include "header-file"><stdio.h>
2  "include">#include "header-file"><stdlib.h>
3
4  "type">int "function">main() {
5      "type">int n, head, seek_time = 0;
6      "function">printf("Enter the number of disk requests: ");
7      "function">scanf("%d", &n);
8      "type">int request_queue[n];
9      "function">printf("Enter the disk request queue:\n");
10     "keyword">for ("type">int i = 0; i < n; i++) {
11         "function">scanf("%d", &request_queue[i]);
12     }
13     "function">printf("Enter the initial position of the disk head: ");
14     "function">scanf("%d", &head);
15
16     // Sort the request queue for simplicity
17     "keyword">for ("type">int i = 0; i < n - 1; i++) {
18         "keyword">for ("type">int j = i + 1; j < n; j++) {
19             "keyword">if (request_queue[i] > request_queue[j]) {
20                 "type">int temp = request_queue[i];
21                 request_queue[i] = request_queue[j];
22                 request_queue[j] = temp;
23             }
24         }
25     }
26
27     // C-SCAN Scheduling
28     "function">printf("\nC-SCAN Disk Scheduling:\n");
29     "type">int start = 0;
30     "type">int end = n - 1;
31     "keyword">while (start <= end) {
32         "keyword">for ("type">int i = start; i <= end; i++) {
33             "keyword">if (request_queue[i] >= head) {
34                 seek_time += "function">abs(head - request_queue[i]);
35                 head = request_queue[i];
36                 start = i + 1;
```

Enter the number of disk requests: **5** Enter the disk request queue: **98 183 37 122 14** Enter the initial position of the disk head: **50** C-SCAN Disk Scheduling: Total Seek Time: 382 Average Seek Time: 76.40

=== Code Execution Successful ===

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37     }
38 }
39 // Move the head to the end in the current direction
40 seek_time += "function">abs(head - 0);
41 head = 0;
42 // Change direction to the opposite side
43 seek_time += "function">abs(head - request_queue[end]);
44 head = request_queue[end];
45 end = n - 2; // Exclude the last request, as it has already been served
46 }
47 "function">printf("Total Seek Time: %d\n", seek_time);
48 "function">printf("Average Seek Time: %.2f\n", (float)seek_time / n);
49 "keyword">return 0;
50 }
```